

- 4** The cubic equation $27x^3 + 18x^2 + 6x - 1 = 0$ has roots α, β, γ .

- (a) Show that a cubic equation with roots $3\alpha + 1$, $3\beta + 1$, $3\gamma + 1$ is

$$y^3 - y^2 + y - 2 = 0. \quad [3]$$

[illegible]

The sum $(3\alpha + 1)^n + (3\beta + 1)^n + (3\gamma + 1)^n$ is denoted by S_n .

- (b) Find the values of S_2 and S_3 . [4]

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- (c) Find the values of S_{-1} and S_{-2} . [3]

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